



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examination 2018
Higher 2

CANDIDATE
NAME

SUGGESTED ANSWERS

CIVICS
GROUP

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INDEX
NUMBER

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H2 BIOLOGY

9744

Practical Exam

28 August 2018
2 hours 30 minutes

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on all the work you hand in.

Give details of the practical shift and laboratory, where appropriate, in the boxes provided.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams and graphs.

Do not use staples, paper clips, glue or correction fluid/tape.

Answer **all** questions in the spaces provided on the Question Paper.

The use of scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

Shift
Laboratory

For examiner's Use	
1	/ 21
2	/ 18
3	/ 16
Total	/ 55

QUESTION 1

Many plant cells contain a water soluble molecule called ascorbic acid, which has many functions to help the plant survive.

You are provided with an extract from plant cells, **P**, which contains ascorbic acid.

Visking tubing, **V**, is partially permeable, similar to a cell surface membrane, so that some biological molecules will diffuse through the wall of the tubing.

You are required to investigate the diffusion of ascorbic acid from **P** into the water surrounding the Visking tubing over a period of 15 minutes.

Fig. 1.1 shows the apparatus before the water was added.

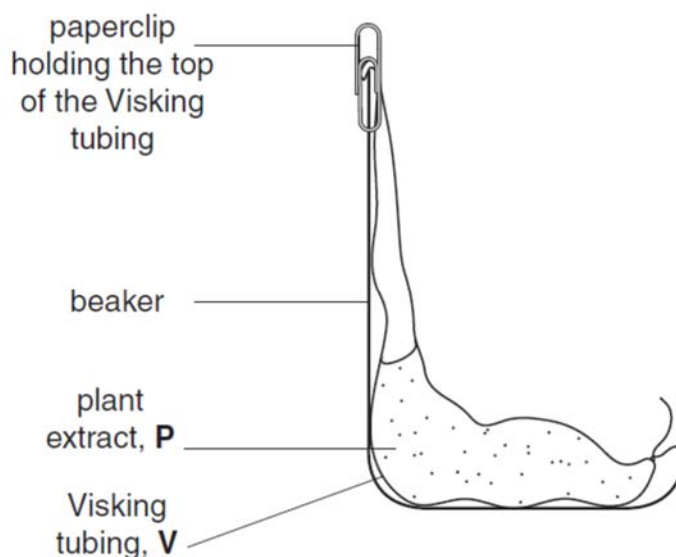


Fig. 1.1

- (a) (i) Water is added to the beaker in Fig. 1.1.
Describe the expected trend in the concentration of ascorbic acid in the water over a period of 15 minutes.

Idea of increase ;

[1]

You are provided with:

labelled	contents	hazard	volume / cm ³
A	sample of water removed after 15 minutes	irritant	15
P	plant extract containing ascorbic acid	irritant	15
W	distilled water	none	100
I	iodine in potassium iodide solution	irritant	20
S	starch	none	20

labelled	contents	hazard	details	quantity
V	Visking tubing	none	18 cm length in a beaker containing water	1

You must now read up to the end of step 23 before proceeding.

To compare the concentration of ascorbic acid in the samples you are required to find the **volume** of iodine solution, **I**, added to each sample until the end-point is reached.

The drops of **I** will be added one at a time using a small syringe.

To practise releasing drops from a small syringe:

1. Fill the syringe with 2 cm³ of **I**.
2. Hold the syringe over an **empty** test-tube and push the plunger **gently** to release one drop at a time, as shown in Fig. 1.2.

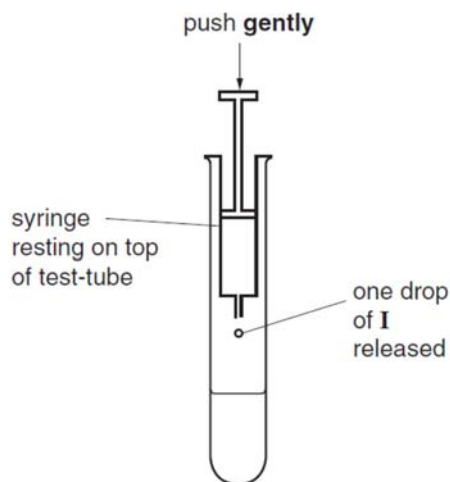


Fig. 1.2

To compare the concentration of ascorbic acid in the samples you will need to add drops of **I** until a blue colour appears. When this blue colour lasts for more than 10 seconds, this indicates the end-point and the **volume of I** that has been added should be recorded.

The apparatus in Fig. 1.1 was set-up and water was added and left for 15 minutes. Sample **A** was removed from the water in the beaker.

You are required to find the volume of **I** needed to reach the end-point for sample **A**.

Proceed as follows:

3. Put 1 cm³ of **S** into a test-tube.
4. Put 3 cm³ of the sample (e.g. **A**) into the same test-tube.
5. Shake the test-tube gently to mix the contents.
6. Fill the syringe, labelled **I**, with 2 cm³ of **I**.
7. Wipe off any drops of **I** from the outside of the syringe with a paper towel.
8. Add **one** drop of **I** to the mixture in the test-tube as shown in Fig. 1.2.
9. Mix gently and if there is no colour change add another drop.
10. Continue adding drops, one at a time, until the blue colour appears.
Wait 10 seconds to see if the end-point has been reached. If the blue colour disappears then add another drop.
11. Repeat step 10 until the mixture stays blue for at least 10 seconds.

(ii) Record the **volume of I** needed to reach the end-point.

volume: 1.2 cm³ (accept range from 0.9 cm³– 1.3 cm³)

[1]

You are required to:

- set up a Visking tubing containing **P** as shown in Fig. 1.3
- decide the level of water to put into the beaker
- remove samples of the water surrounding the Visking tubing at **5 minute intervals** for 15 minutes
- compare the ascorbic acid concentrations in the samples.

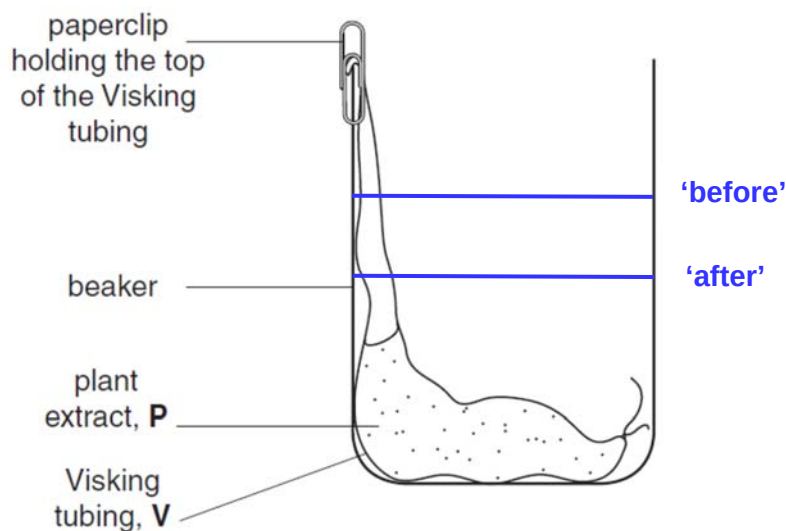


Fig. 1.3

- water level must still cover contents of Visking tubing ;
- water level must be lower than 'before' ;

Samples of water surrounding the Visking tubing will be removed for testing, so you need to take this into account when you decide the level of water to put into the beaker.

(iii) Draw **on Fig. 1.3** the level of the water

- before you remove any samples (label 'before'),
- after the total volume of water needed for all the tests has been removed (label 'after').

two levels drawn labelled with 'before' + 'after' + water level after lower than before [1]
lowest level still covers contents of Visking tubing [1]
[2]

(iv) In order to compare the ascorbic acid concentrations, state **one** variable which you will need to standardise when finding the volume of **I** added to each sample.

Describe how you will standardise this variable.

variable: volume of ascorbic acid / starch

description: use the same syringe to draw 3 cm³ of ascorbic acid / 1 cm³ of starch;

Both must be given [1]

Proceed as follows:

12. Put **S**, as in step 3, into the four test-tubes you will require in order to test the samples of water.

13. Tie a knot in the Visking tubing as close as possible to one end so that it seals the end.

14. To open the other end, wet the Visking tubing and rub the tubing gently between your fingers.
15. Put 6 cm³ of **P** into the open end of the Visking tubing.
16. Rinse the outside of the Visking tubing by dipping it into the water in the container labelled **V**.
17. Put the Visking tubing into an empty beaker as shown in Fig. 1.3.
18. Make sure the open end of the Visking tubing is held in place by a paperclip.

You will start timing as soon as you add **W** (steps 19 and 20).

You should read steps 19 to 23 before proceeding.

19. Put **W** into the beaker to the level you decided in (iii).
20. Immediately start timing and remove the first sample of water (as in **step 4**) and put into a prepared test-tube (as in **step 12**).
21. Test the sample as in **steps 5 to 11**.
22. After 5 minutes, gently mix the water surrounding the Visking tubing and then remove the next sample, put it into a different (prepared) test-tube and repeat **steps 5 to 11**.
23. Repeat **step 22** for two more samples.

Record your results in **(a)(v)** on page 7.

v) Record your results in the space below.

Time / min	Volume of iodine / cm ³
0	0.1
5	0.3
10	0.8
15	1.5

All grid lines of the table drawn [1]
 Correct heading with unites [1]
 Records results at four times (0, 5, 10, 15) [1]
 increasing trend of volume of iodine [1]
 All values to one decimal place [1]
[5]

(vi) Describe how the results support your expected trend as stated in **(a)(i)**.

Result showed increasing trend of volume of iodine , which supports the expected increasing trend stated in (a) (i)

[1]

This investigation provides results to compare the concentration of ascorbic acid in the samples.

- (vii) If you had been provided with 1.0% ascorbic acid solution, suggest how you would modify this investigation to find the **percentage concentration** of ascorbic acid in the water after 15 minutes.

1. **[Compulsory]** Idea of serial dilution or simple dilution (of 1.0% ascorbic acid) ;
2. **[Any 1]** Idea of : Use standard graph of known ascorbic acid concentrations to find % concentration of ascorbic acid in the water after 15 mins ;
 - Record the volume of iodine solution needed to reach end-point for each known concentration of ascorbic acid. Plot the standard graph.
 - Use the standard graph to find the % concentration of ascorbic acid in the water after 15 mins.

[2]

- (b) Iodine solution (iodine in potassium iodide solution) turns blue-black when starch is present in plant tissues.

However, as ascorbic acid is also found in plant tissues, some scientists investigated the effect of testing for starch with iodine solution when there was ascorbic acid present.

The concentration of ascorbic acid was $0.0001 \text{ mol dm}^{-3}$ and the concentration of starch solution was standardised.

The percentage of starch which reacted with the iodine solution was measured.

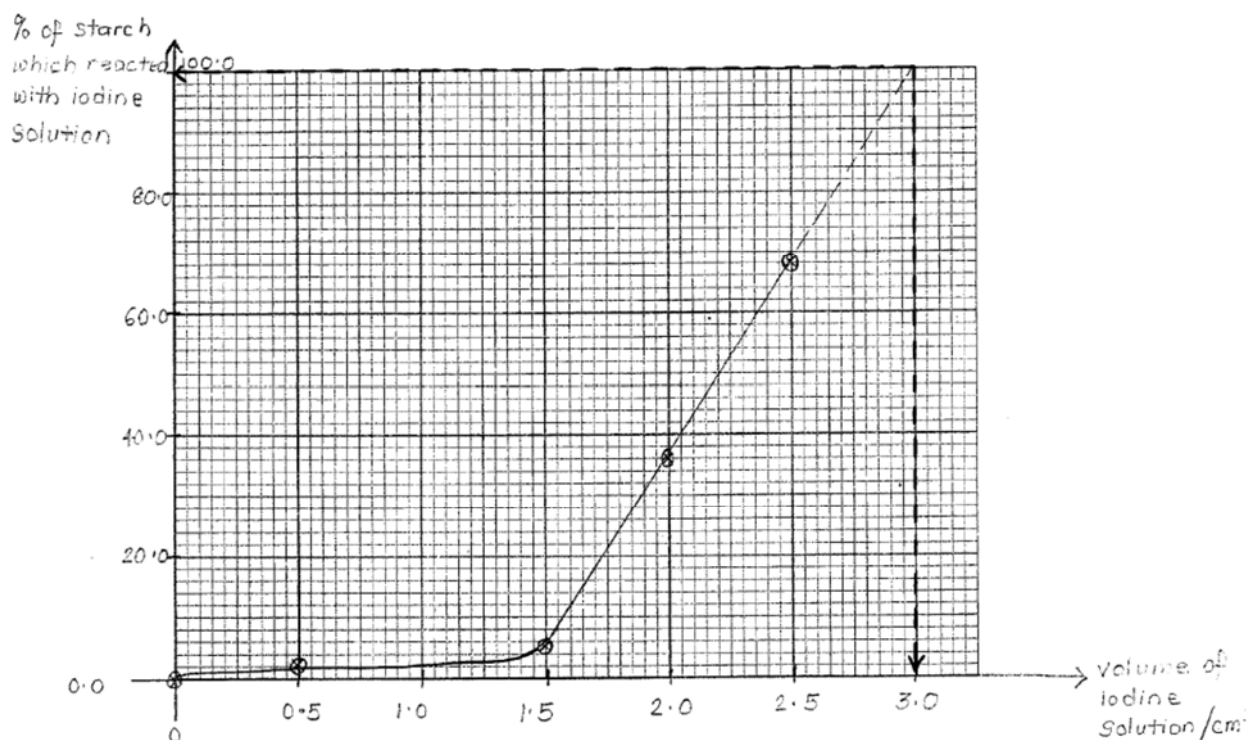
The results are shown in Table 1.1.

Table 1.1

volume of iodine solution / cm^3	percentage of starch which reacted with iodine solution / %
0.0	0.0
0.5	2.0
1.5	5.0
2.0	36.0
2.5	68.0

(i) Plot a graph of the data shown in Table 1.1.

You will need to consider the answer to (b)(ii) before you plot your graph.



Correct labelling of x and y axes with correct units [1]
 Appropriate scale plotted (at least 2/3 of the space provided) [1]
 Correct plotting of five points as small cross or dot in circle or cross [1]
 Five plots + point-to-point or best fit graph [1]

[4]

(ii) Estimate the volume of iodine solution needed for 100% of the starch to be reacted.
 Show on your graph how you obtained the volume of iodine solution.

volume of iodine solutione.g. 3.0..... cm³

Marking point:

Correct estimation from graph by shown extrapolation ;

[1]

(iii) Explain how the presence of ascorbic acid may affect the use of iodine solution as a test for the presence of starch in different plant tissues.

1. Idea of too much ascorbic acid then iodine may not stain or idea of having to add more iodine in order to observe colour ;

2. Need to know how much ascorbic acid in plant tissues to make test accurate ;

[2]

- (iv) A plant tissue contains starch and $0.0001 \text{ mol dm}^{-3}$ ascorbic acid.
Suggest how you would make sure that the iodine test showed the presence of all the starch (100%).

Use 3 cm^3 or more of iodine or more iodine, or excess iodine OR

The volume given in (b) (ii) or more ;

[1]

[Total: 21]

QUESTION 2

Some plants store starch as grains.

Some starch grains have rings on the surface and the grains from different plant types differ in size and shape.

You are required to:

- observe and draw starch grains from two different types of plant
- compare the starch grains from these two different types of plant.

You are provided with starch grains, in water, from two different types of plant, labelled **F** and **G**.

1. Put one **clean and dry** microscope slide on a piece of black card with a paper towel underneath as shown in Fig. 2.1.

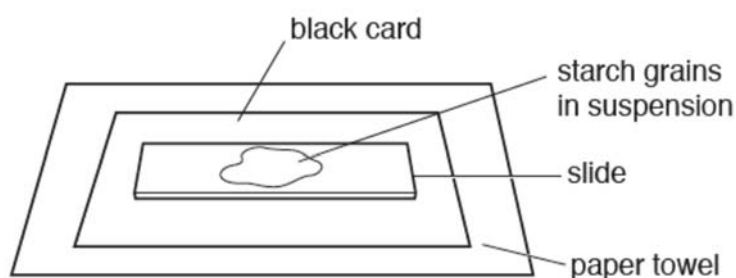


Fig. 2.1

2. Using a pipette, put a few drops of **F** containing starch grains onto the slide.
3. Cover the starch grains with a coverslip and use a paper towel to remove any excess liquid that is outside the coverslip.
4. View the slide to observe the starch grains using the microscope. Select an area of starch grains on the slide.

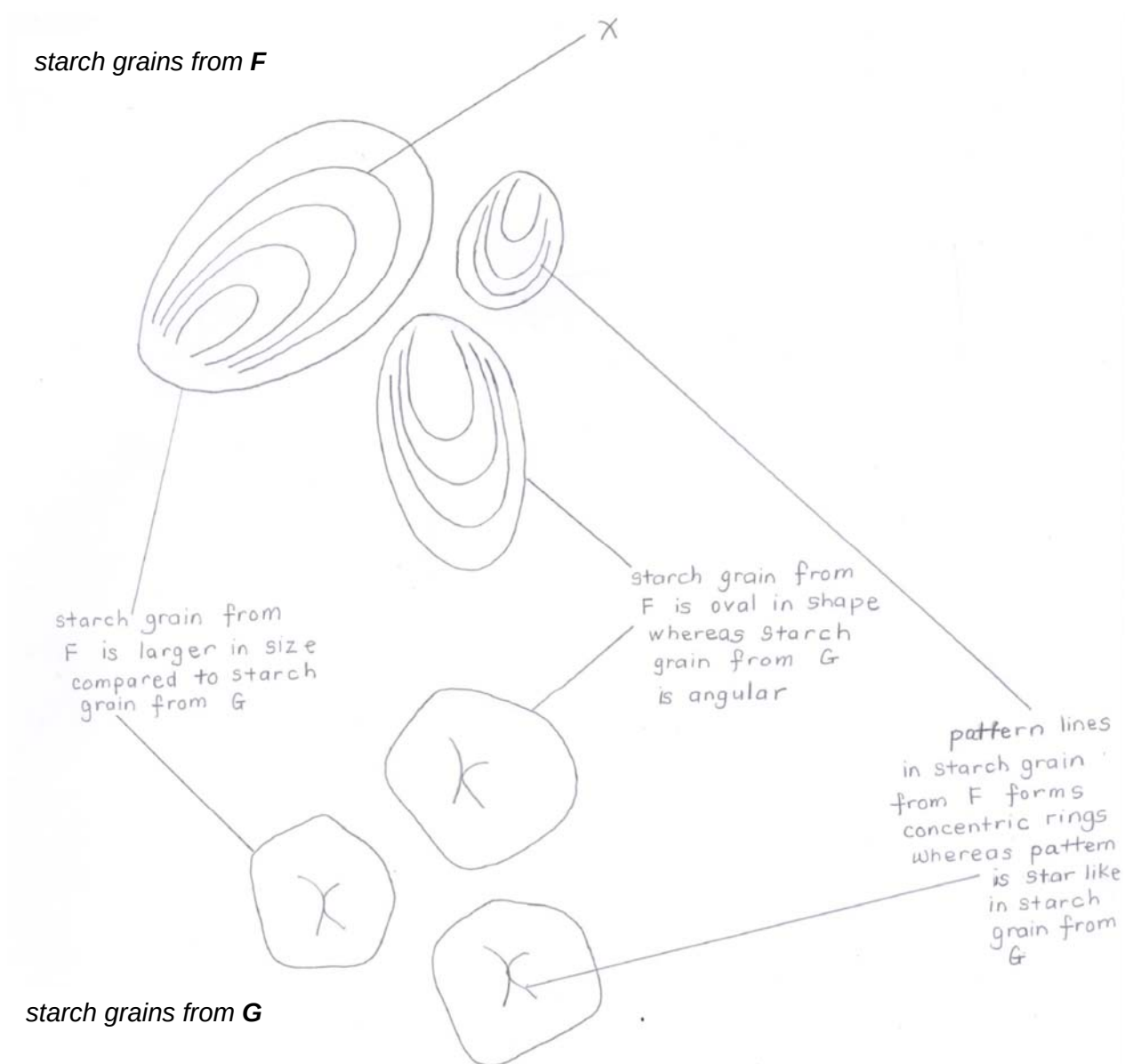
You may need to reduce the amount of light entering the microscope to observe the grains clearly.

You must adjust the fine focus to observe the starch grains.

5. Select **three** starch grains which show the different sizes and features of the grains.
6. Make a large drawing in **(a)(i)** of the three starch grains that you have selected.
7. Remove the slide from the microscope and place on a paper towel.
8. Repeat step 1 to step 7 with **G**.

Use a sharp pencil for drawing.

- (a) (i) Make a large drawing of the three starch grains from **F** and the three starch grains from **G**.
Use **one** ruled label line and label **X** to identify the surface markings of **one** starch grain.



Quality of line (thin and continuous) for at least 6 grains drawn [1]

Draws only 3 starch grains from F + only 3 starch grains from G + grains not overlapping [1]

Starch grains from F drawn as oval shapes + starch grains from G drawn as angular shapes [1]

Uses one label line + one label, X, to identify surface markings on grains [1]

[4]

(ii) Annotate your drawings in (a)(i) to describe **three** observable differences between the starch grains from **F** and **G** by:

- drawing label lines to each of the features on the starch grains that show these differences
- describing next to each line how each feature is different.

[1 mark for each annotated difference]

The eyepiece graticule scale in your microscope may be used to measure the actual length of the layers of tissue or cells, if the scale has been calibrated against a stage micrometer.

You are provided with:

- Microscope with an eyepiece graticule fitted into the eyepiece lens
- Stage Micrometer: 1 mm divided into 100 divisions. Each is 0.01mm.

(b) (i) Using the appropriate objective lens and the eyepiece graticule, find the actual width, in μm , of the three starch grains that you have drawn from **F**

You may lose marks if you do not show your working.

Working for the calibration of the eyepiece graticule scale for the appropriate objective lens that you have selected:

Example:

Under the x40 objective lens:

100 eyepiece graticule divisions = 26 (A: 24-27) stage micrometer divisions

1 eyepiece graticule division

= 26 divided by 100 stage micrometer divisions

= 26 divided by 100 $\times$ 0.01 mm

= 26 divided by 100 $\times$ 0.01 $\times$ 1000 μm

= 2.6 μm

Working for the measurement of the starch grains that you have drawn from **F**:

Example:

Under the x40 objective lens:

Actual width of the 1st starch grain

= 29 eyepiece graticule divisions

= 29 $\times$ 2.6 μm

= 75.4 μm

Actual width of the 2nd starch grain

= 27 eyepiece graticule divisions

= 27 $\times$ 2.6 μm

= 70.2 μm

Actual width of the 3rd starch grain
= 15 eyepiece graticule divisions
= 15 x 2.6 μm
= 39.0 μm

Chosen objective lens stated + calibration of 100 EP graticule divisions [1]

Correct calibration of one EP graticule division in mm / μm [1]

Working shown clearly throughout [1]

1 d.p. given for μm [1]

Correct final measurements of starch grains in μm [1]

actual width of starch grains: 1..... 2..... 3..... [5]

(ii) Using the actual widths calculated in (i), calculate the **mean** actual width of a starch grain that you have drawn from F.

You may lose marks if you do not show your working or if you do not use appropriate units.

Mean actual width of a starch grain that you have drawn from F

= [75.4 + 70.2 + 39.0] divided by 3 μm ; [1]

= 61.5 μm ; correct calculated value given to one d.p / whole number [1]

mean actual width[2]

Fig. 2.2 is a photomicrograph of a stained transverse section through part of a leaf of the plant from which starch grains labelled **G** were taken from.

You are not expected to be familiar with this specimen.

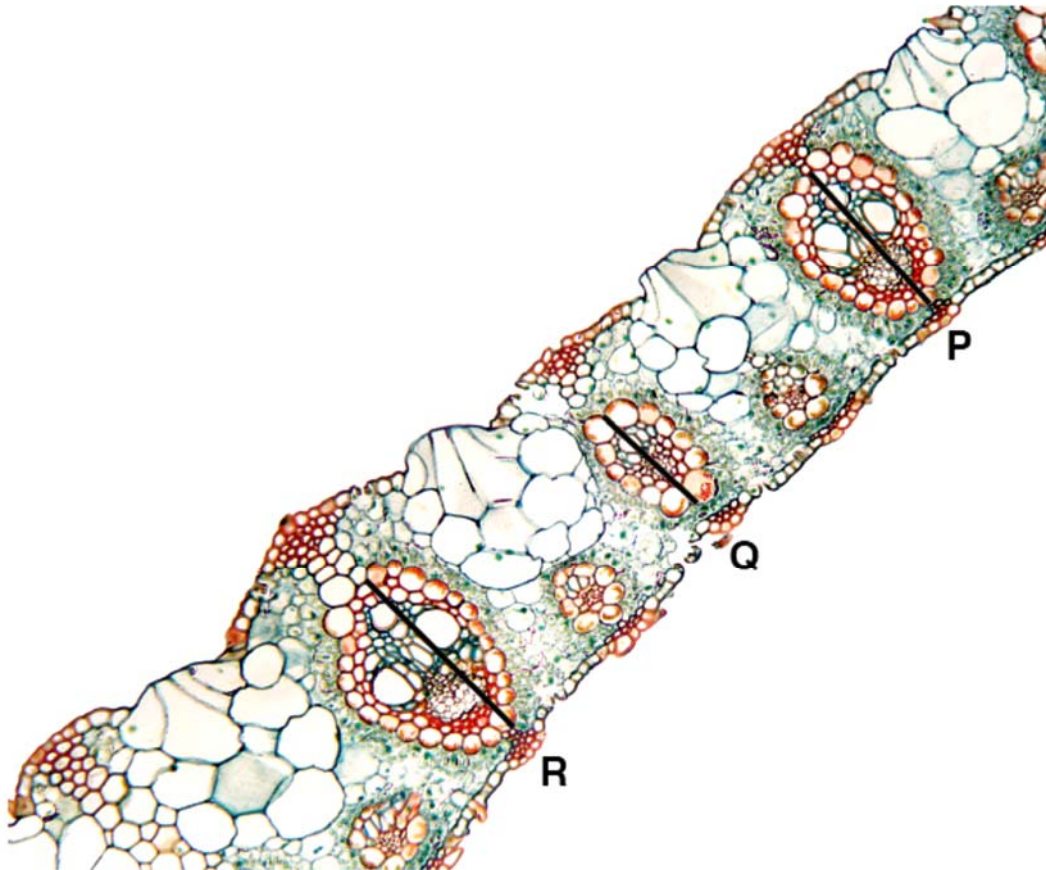


Fig. 2.2

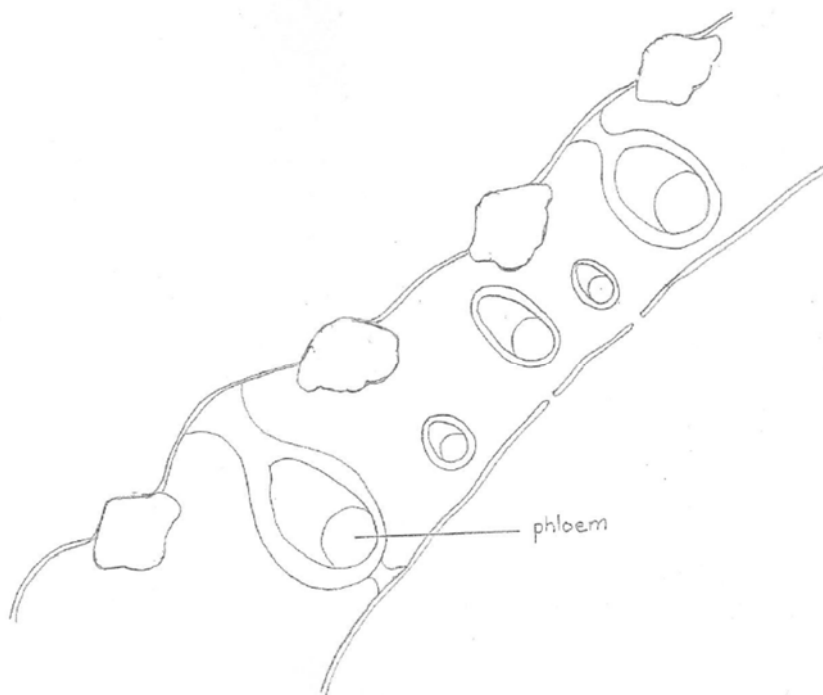
Observe the different tissues in Fig. 2.2 and select an area that shows part of the epidermis and the vascular bundles.

Use a sharp pencil for drawing.

(c) Draw **one** large plan diagram from the selected area which shows:

- part of the epidermis
- only **three** larger vascular bundles labelled **R**, **Q** and **P**
- only **two** smaller vascular bundles
- any other observable tissues.

Use **one** ruled label line and label to identify the phloem.



[4]

Compulsory marking points:

1. correct section drawn + draws only 3 large vascular bundles (R, Q and P) and 2 smaller vascular bundles in the correct position and correct proportion [1]
2. draws at least one vascular bundle divided into at least 3 sections + uses one label line + one label to identify phloem [1]

Any two marking points from the following:

3. plan diagram of appropriate size (at least 11 cm) + no cells + no shading [1]
4. draws the region showing the 'bulliform cells' / stoma [1]
5. draws epidermis as 2 lines [1]

[Total: 18]

Note:

For those candidates who started with Question 2, start Question 1 as soon as you have completed Question 2.

QUESTION 3

The information on the label of a bottle of one brand of cleaning solution for contact lenses states that it contains enzymes which remove protein from contact lenses.

The instructions for using the solution state that:

- The cleaning solution should be poured into the cleaning pot provided up to the marked level.
- The contact lenses should be left in the cleaning solution for a minimum of four hours.
- Before using the lenses, they should be washed in a sterile saline (sodium chloride) solution.

The contents of the solution are:

1. Subtilisin A, a protease with an optimum temperature of 60 °C and an optimum pH of 7.0–7.5.
2. Buffered saline, a physiological solution with the same 'balance of ions' as body fluids.
3. Disodium EDTA, a preservative.

A student was asked to find out the actual concentration of the protease present in the contact lens cleaning solution.

The student found a web site that gave the concentration of subtilisin A in different brands of contact lens cleaning solutions. These range between $20 \mu\text{g cm}^{-3}$ and $100 \mu\text{g cm}^{-3}$.

The student simulated a dirty contact lens using the protein gelatin and a thin transparent plastic sheet. One side of the plastic sheet was dipped into melted gelatin containing a dye. The gelatin was then allowed to set.

Fig. 3.1 shows a simulated contact lens.

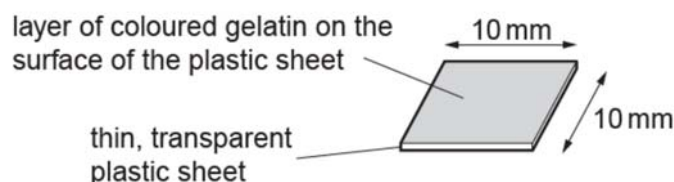


Fig. 3.1

The student tested the activity of subtilisin A by recording the time taken for the coloured gelatin to be removed, leaving the transparent plastic sheet.

Fig. 3.2 shows the apparatus the student used.

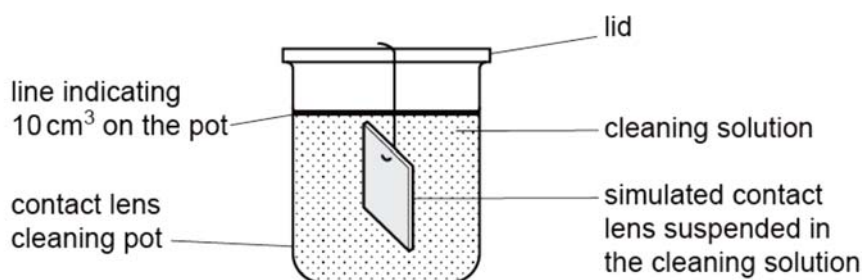


Fig. 3.2

Table 3.1 shows the results of trials to find the time taken for the coloured gelatin to be removed by the cleaning solution at 35 °C.

Table 3.1

trial	1	2	3
time / minutes	35	50	40

- (a) The student noticed that there was a large variation in the results. This may be because it was difficult to know when to stop the stopwatch.

Suggest two problems in deciding when to stop the stopwatch in this investigation.

Expected answer		
Correct independent variable given ; [1] Correct dependent variable given ; [1] 1. Independent variable: concentration of subtilisin A solution / unit, range of 5 uniformly spaced concentrations (e.g. 20 / 40 / 60 / 80 / 100 $\mu\text{g cm}^{-3}$ or 0.02 / 0.04 / 0.06 / 0.08 / 0.1 mg cm^{-3}) 2. Dependent variable: time for the disappearance of gelatin or colour / s or rate of disappearance of gelatin or colour + unit / s^{-1} ;		
any 2 from : [2] 3. dilution of stock solution (1mg cm^{-3} solution) 10 times to give 100 $\mu\text{g cm}^{-3}$ solution ; 4. correct method of dilution as shown in dilution table / proportional dilution ; Reject dilution with water alone		
Concentration of subtilisin / $\mu\text{g cm}^{-3}$	Volume of 100 $\mu\text{g cm}^{-3}$ or 0.1 mg cm^{-3} subtilisin A & buffered saline solution / cm^3	Volume buffered saline solution / cm^3
20	10.0	40.0
40	20.0	30.0
60	30.0	20.0
80	40.0	10.0
100	50.0	0.0

5. total volume of 50 cm³ ;
Correct Design of Control experiment and Purpose given [1] 7. [Design] Boiled cleaning solution (denatured subtilisin A) [Purpose] To show that subtilisin A digests / remove the gelatin layer ; Or [Design] Buffered saline solution without subtilisin A [Purpose] To show that other components of the cleaning solution do not digest / remove the gelatin layer ; Reject distilled water
Expected answer
<i>any 6 from the following [6]</i> independent variable 8. ref. to using 10 cm³ of each concentration of the subtilisin A solution ; 9. method of measuring volume ; e.g. graduated pipette / syringe / measuring cylinder / burette dependent variable: 10. incubate the subtilisin A solution to equilibrate the test temperature (35°C) for minimum of 2 mins ; 11. use, stopwatch / timer, to record end point ; standardising variables [max 3] • stating any 2 correct variables [1] • stating how variable 1 is controlled [1] stating how variable 2 is controlled [1] 12. stating any 2 correct variables i.e. temperature / pH; 13. ref. to method of keeping incubation temperature controlled e.g. thermostatically controlled water bath set at 35°C ; 14. idea of standardising the (coloured) gelatin (thickness / coverage / distribution) – specify instrument used for measurement of thickness ; 15. use of, buffer to keep pH (7.0–7.5) constant ; 16. ref. to using the same size / area, of (simulated) contact lens e.g. 10 mm × 10 mm or specify instrument used for measurement ;
Correct risk assessment & safety precaution given: [1]

17. reference to risk and suitable safety precaution ;

[Risk] Subtilisin A or disodium EDTA is an irritant

[Safety precaution] Wearing goggles and gloves

Reject: no risk and no safety implications

Reject: Subtilisin A or disodium EDTA is corrosive

Consideration of Reliability [1]

17. ref. to minimum number of **at least three replicates** and finding the **mean** to minimize errors/**ensure accuracy** and **repeating at least two more times using fresh materials** and freshly prepared solutions / new batch of materials and solutions to **ensure reproducibility** of trend and reliability of results ;

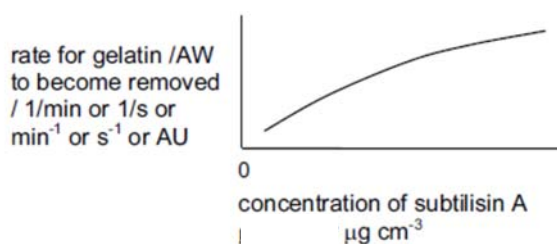
Expected answer

18. Table of results [1]

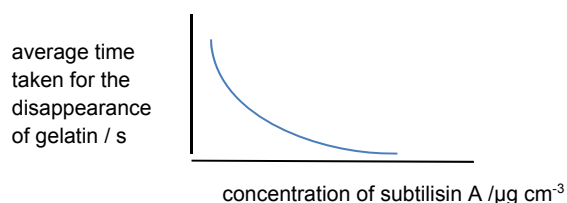
Concentration of subtilisin / $\mu\text{g cm}^{-3}$	Time for disappearance of gelatin / s or min				Rate for gelatin to be removed / s^{-1} or min^{-1} or AU
	R1	R2	R3	Average	
20					
40					
60					
80					
100					

Accept with or without computation of rate

19.



Also accept – Plot of the average time taken for the disappearance of gelatin against concentration of subtilisin A



20. idea of:

find the time for the gelatin to disappear (using the cleaning solution) on the y-axis and read the concentration ($*C_1$) from the x-axis ;

21. idea of:

repeating using narrower intervals of concentration of subtilisin A [within range close to ($*C_1$)]

22. Rinse plastic sheets (previously dipped into melted gelatin containing a dye and allowed to set) provided with distilled water before using for the experiment.

[Total: 16]

THE END